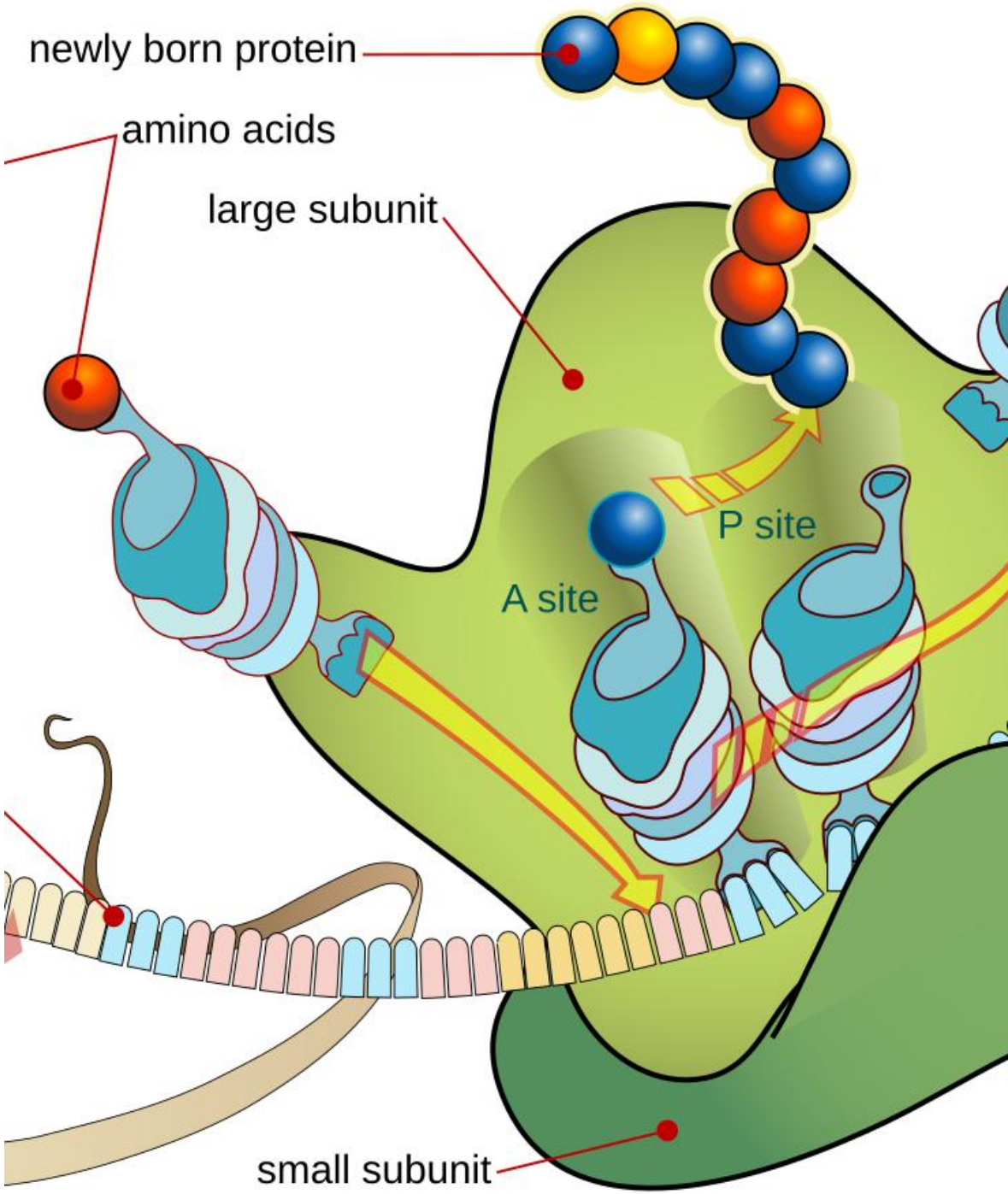
DP Biology – D1.2 (HL extension)

Inside protein synthesis

- How does a cell turn a DNA base sequence into a chain of amino acids?
- What keeps protein synthesis reliable?

HL statements D1.2.12–19. Textbook pp. 625–636.
Image: a ribosome translating mRNA. LadyofHats, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D1.2.12–14	Explain the direction of transcription and translation, how transcription starts, and what non-coding DNA does
2	D1.2.15–17	Explain mRNA processing and alternative splicing, and how translation starts at the A, P and E sites
3	D1.2.18–19	Explain how polypeptides are modified, using insulin, and how proteasomes recycle proteins

Direction in transcription and translation

D1.2.12

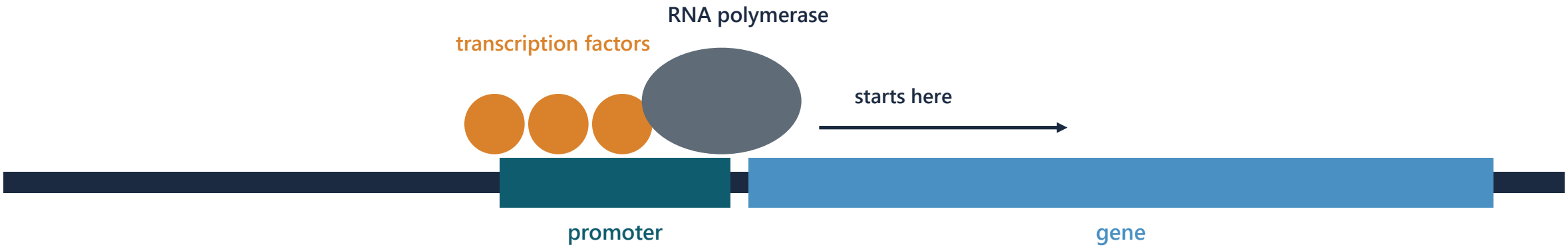
	Transcription	Translation
Done by	RNA polymerase	Ribosome
Reads	Template DNA, 3′ to 5′	mRNA, 5′ to 3′
Builds	RNA, 5′ to 3′	Polypeptide, N end to C end
Why	Nucleotides join the 3′ -OH	Start codon is near the 5′ end

Key idea

A fixed direction means the same base sequence always gives the same amino acid sequence.

Starting transcription at the promoter

D1.2.13



Transcription factors bind the **promoter**; RNA polymerase joins them to form an initiation complex. Then it separates the strands and starts making RNA.

Without the right factors, a gene is not transcribed.

Common mistake

RNA polymerase is not a transcription factor: the factors help it bind.

DNA that does not code for polypeptides

D1.2.14

Sequence	Role
Regulators	Promoters, enhancers, silencers: where transcription factors bind
Introns	Inside genes; removed from mRNA
Telomeres	Repeats at chromosome ends that protect genes during replication
rRNA and tRNA genes	Transcribed but not translated: the RNA is the product

Scale
Only about 1.5% of the human genome codes for protein.

Practice: direction and control

D1.2.12–14

Ends

Label the 5′ and 3′ ends of an mRNA and the N and C ends of its polypeptide.

No factor

A cell lacks one transcription factor. Predict the effect.

Telomeres

Suggest what happens to chromosome ends without telomeres.

Explain

Give a reason for each answer.

Working with data: what the genome contains

D1.2.14

Approximate make-up of the human genome (published estimates, rounded).

Type of sequence	% of genome
Protein-coding exons	1.5
Introns	25
Regulatory sequences	about 8
Repeated sequences	about 50
Other	about 15

1. Calculate the non-coding percentage.
2. The genome is 3 billion base pairs. Estimate the protein-coding bases.
3. Explain why non-coding is not the same as useless.

Check-in: class 1

D1.2.12–14

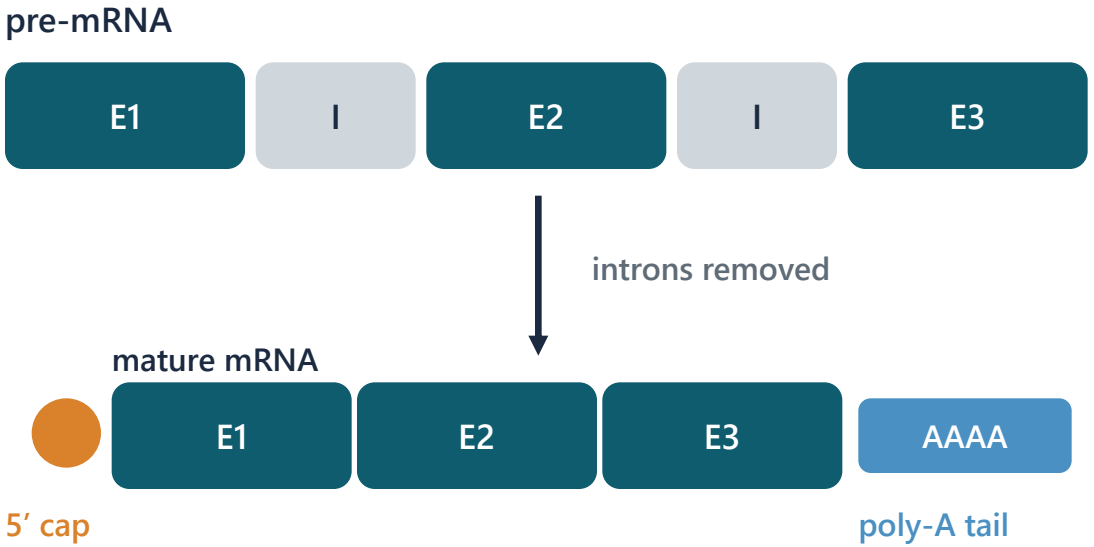
1. State the direction RNA is built in transcription.
2. State which end of the mRNA the ribosome reads first.
3. Explain the role of the promoter.
4. Give three kinds of non-coding DNA.
5. Explain why a cell can switch a gene off.

How to use this

Answer without your notes, then check against slides 3–7.

Processing mRNA in eukaryotes

D1.2.15



Change	Purpose
Introns out, exons spliced	Only coding sequence left
5′ cap	Protects; helps the ribosome bind
Poly-A tail	Protects; helps export

Where and why

This happens in the nucleus. Prokaryotes have no introns, so to make a human protein in bacteria an intron-free copy of the gene (made from mRNA) is used.

One gene, several proteins

D1.2.16



Exons of the troponin T gene (simplified)



Foetal heart form

Adult heart form

So

Alternative splicing keeps or leaves out particular exons, giving different polypeptides (isoforms) from one gene. About 20 000 human genes give far more kinds of protein.

Practice: how many proteins?

D1.2.16

A gene has 5 exons. Exons 1 and 5 are always kept; exons 2, 3 and 4 can each be kept or left out.

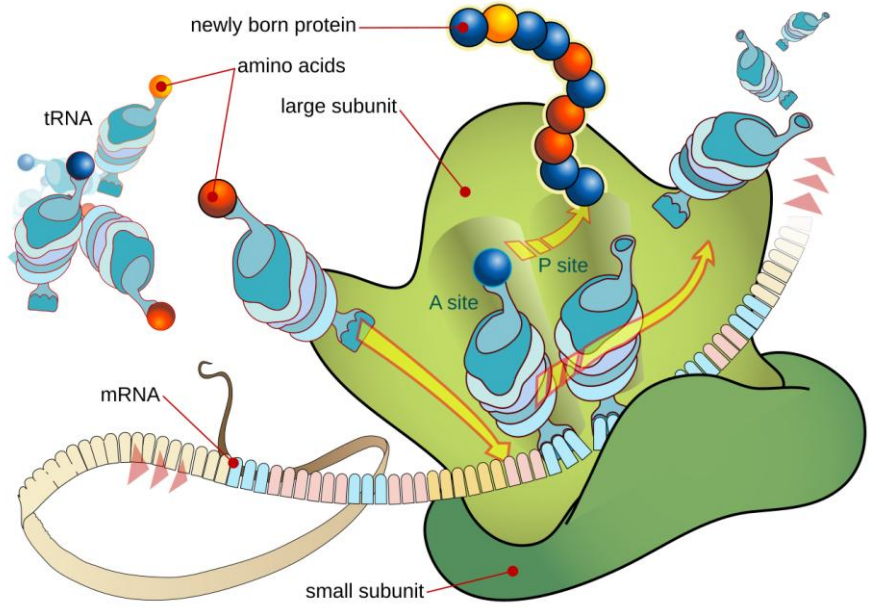
1. Calculate how many different mRNAs this gene could produce.
2. Humans have about 20 000 protein-coding genes but make over 80 000 proteins. Explain how.
3. Suggest why splicing must be very accurate.

Getting ready to translate

D1.2.17

A specific **tRNA-activating enzyme** attaches each amino acid to the tRNA with the matching anticodon, using ATP.

The **initiator tRNA** carries methionine and has anticodon UAC. The large subunit has three sites: **A**, **P** and **E**.



Ribosome with A and P sites. LadyofHats, public domain.

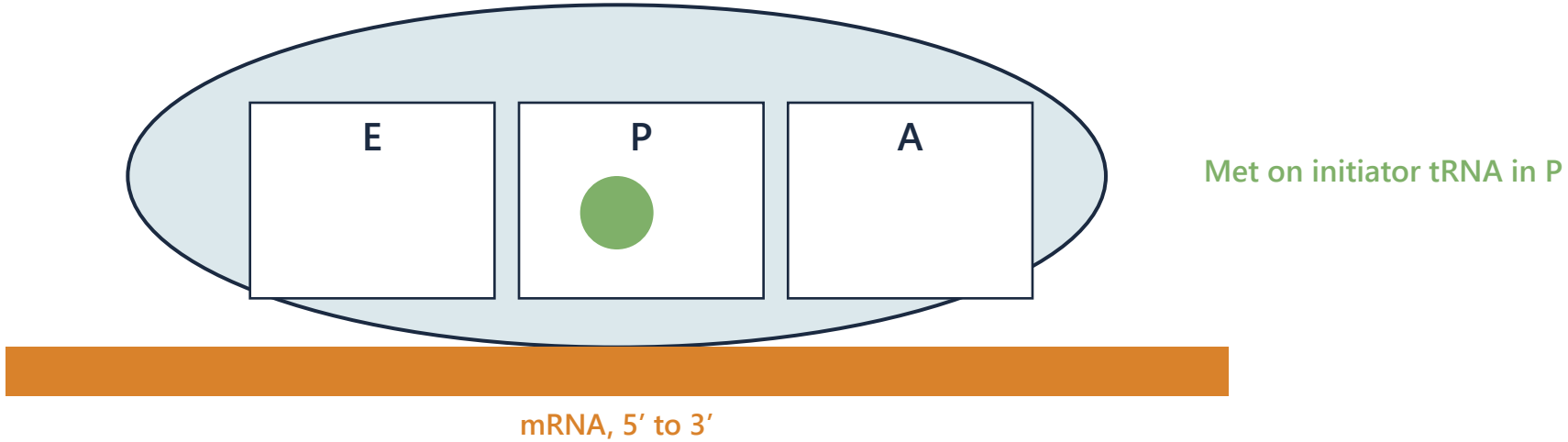
Accuracy

Each activating enzyme recognises one amino acid and its tRNAs, so the right amino acid matches each anticodon.

Initiation of translation

D1.2.17

Small subunit binds the 5′ end of mRNA → Moves to the start codon AUG → Initiator tRNA (Met) pairs with AUG → Large subunit joins: tRNA in the P site



Met on initiator tRNA in P

The A, P and E sites in elongation

D1.2.17

Site	Role in each cycle
A	tRNA with the next amino acid arrives and pairs with the codon
P	Holds the tRNA with the growing chain; the chain moves onto the A-site amino acid: a peptide bond
E	Empty tRNA leaves after the ribosome moves on one codon

Ending

At a stop codon a release factor binds the A site, and the polypeptide is released.

Act out translation

D1.2.17

1. Use cards for codons, tRNAs and amino acids, and three chairs for A, P and E. Translate AUG GCU UUC GGA UAA.
2. **Explain** why translation stops at UAA.
3. **Predict** the effect if AUG were changed to AUC.

Check

The growing chain is always attached to the tRNA in the P site.

Practice: processing and translation

D1.2.15–17

Bacteria

Why can’t bacteria make a human protein from the human gene with introns?

Tail

Predict the effect of a shorter poly-A tail.

Sites

A tRNA in the E site still holds its amino acid. Is this possible?

Explain

Give a reason for each answer.

Check-in: class 2

D1.2.15–17

1. Describe three changes made to pre-mRNA.
2. Explain how alternative splicing gives different proteins.
3. State the role of tRNA-activating enzymes.
4. Outline how translation starts.
5. State the roles of the A, P and E sites.

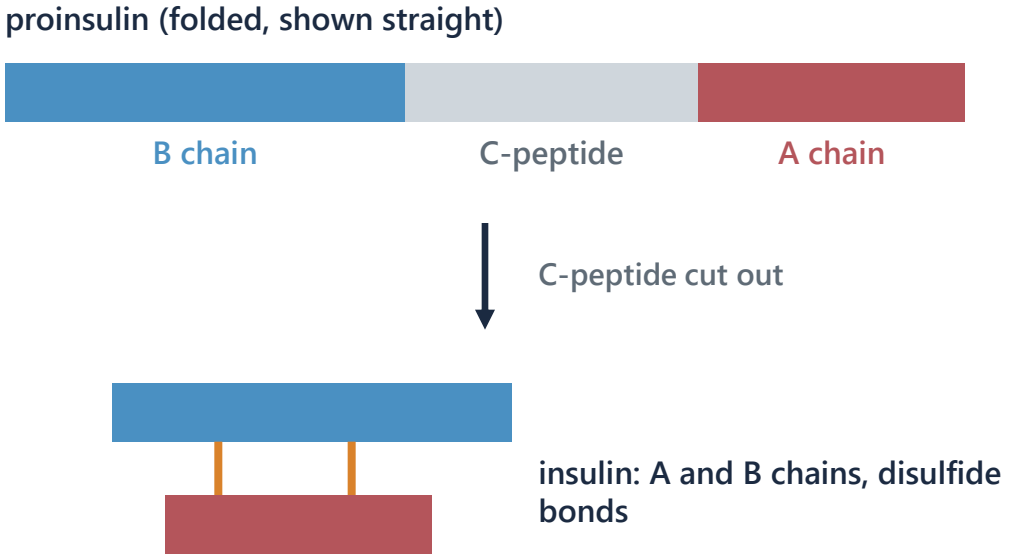
How to use this

Answer without your notes, then check against slides 9–16.

From pre-proinsulin to insulin

D1.2.18

Pre-proinsulin: 110 amino acids, on rough ER → Signal sequence cut off: proinsulin → Folds; three disulfide bonds form → C-peptide cut out in vesicles → Insulin: A and B chains

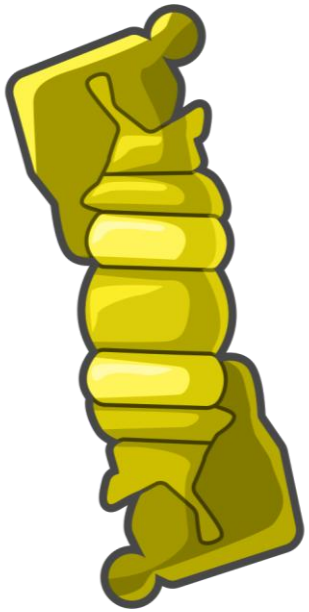


Making insulin as an inactive precursor stops it acting inside the cell that makes it.

Other modifications: cutting, folding, joining chains, adding sugars or phosphate.

Recycling proteins: proteasomes

D1.2.19



Damaged, misfolded or no longer needed proteins are tagged with **ubiquitin**. A **proteasome** unfolds and breaks them into short peptides; the amino acids are reused.

A working proteome needs breakdown as well as synthesis.

A proteasome. NIAID, public domain.

Why?

Proteins controlling the cell cycle must be removed on time; damaged proteins could clump and harm the cell.

Practice: after translation

D1.2.18–19

No cutting

Proinsulin is not cut. Predict the effect.

Blocked proteasome

A drug blocks proteasomes. Predict two effects.

Quaternary

Name a protein whose chains must join after translation.

Explain

Give a reason for each answer.

Working with data: protein lifetimes

D1.2.19

Typical half-lives of proteins in human cells (approximate, rounded).

Protein	Half-life
Cell-cycle regulator (cyclin)	minutes
Typical enzyme	1–2 days
Haemoglobin	about 120 days
Collagen	years

1. Suggest why cyclins are broken down fast.
2. Suggest why collagen lasts so long.
3. Explain how a cell lowers the amount of an enzyme quickly.

Check-in: class 3

D1.2.18–19

1. Outline the steps from pre-proinsulin to insulin.
2. Explain why insulin is made inactive first.
3. Give two other kinds of post-translational modification.
4. Describe how proteasomes recycle proteins.
5. Explain why protein breakdown is needed.

How to use this

Answer without your notes, then check against slides 18–21.

Common mistakes to avoid

Not this	But this
Introns are translated.	Introns are removed before translation.
All DNA codes for proteins.	Most human DNA does not code for polypeptides.
Insulin is made in its active form.	It is made as pre-proinsulin and cut.
Ribosomes read mRNA from the 3′ end.	They read it 5′ to 3′.

Summary: protein synthesis at HL

D1.2.12–19

	Key idea
D1.2.12	RNA built 5′ to 3′; mRNA read 5′ to 3′
D1.2.13	Transcription factors at the promoter start transcription
D1.2.14	Non-coding DNA: regulators, introns, telomeres, RNA genes
D1.2.15–16	Splicing, cap, tail; alternative splicing
D1.2.17	Initiation at AUG; A, P and E sites
D1.2.18	Insulin: cut and folded after translation
D1.2.19	Ubiquitin tags; proteasomes recycle

Check that you can...

1. **Explain** the direction of transcription and translation
2. **Outline** the role of the promoter and of non-coding DNA
3. **Describe** mRNA processing and alternative splicing
4. **Explain** initiation of translation and the ribosome sites
5. **Outline** post-translational modification, using insulin
6. **Outline** the role of proteasomes

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State the roles of the A and P sites of a ribosome.
2. Outline two changes made to mRNA before it leaves the nucleus.
3. Explain why insulin is made as pre-proinsulin first.

Image credits

Ribosome mRNA translation en.svg – LadyofHats – Public domain

Proteasome (NIH BioArt 667 – 758780).svg – Courtesy of NIAID Ryan Kissinger – Public domain

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